

Spectral World Renderer: Towards Verifiable 3D Hyperspectral Unmixing

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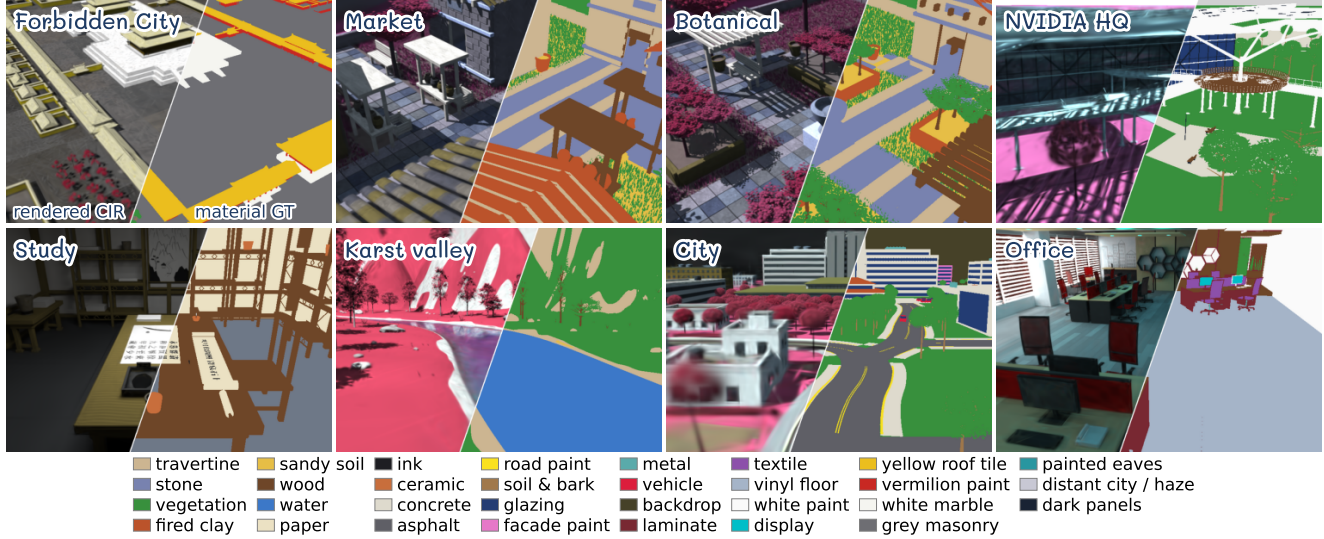


Figure 1. **Hyperspectral worlds for 3D material analysis.** Spectral World Renderer generates eight indoor and outdoor worlds with aligned spectral observations and material composition. Each scene pairs near-infrared imagery rendered by Abundance-GS (left; vegetation appears red) with ground-truth material labels (right), illustrating scene-scale spectral rendering and material analysis.

Abstract

Semantic understanding of 3D scenes does not establish their material composition. Hyperspectral imaging offers material-specific spectral evidence, but costly acquisition, limited composition annotations and expensive multi-view unmixing hinder scene-scale analysis. We present Spectral World Renderer, a framework that couples verifiable hyperspectral simulation with efficient 3D material decomposition. A coding agent translates reference images and text into executable scene programs that specify geometry, materials, illumination and cameras. Measured spectra and shared rendering samples then yield aligned hyperspectral observations, material labels and subpixel material proportions. The resulting SWR-Bench comprises eight indoor and outdoor scenes with material and illumination variation. Within this framework, Abundance-GS embeds a shared endmember dictionary and local mixture coefficients in a compact Gaussian representation, enabling joint spectral rendering and spatial decomposition queries. An appearance-guided learning strategy uses spectral shape and brightness to establish initial endmember

assignments and regularizes their allocation throughout training, without material supervision. Experiments demonstrate improved material identification and spectral fidelity, reduced storage requirements, and approximately 4–10× faster training than the compared pipelines. Together, these contributions enable efficient multi-view material analysis and distinguish spectral reconstruction quality from the validity of its decomposition.

1. Introduction

Understanding a 3D scene requires both semantic recognition and material characterization. Foundation models such as CLIP, SAM and DINO support recognition and spatial queries through transferable features and object masks [30, 33, 49, 57], but these representations do not determine material composition. Hyperspectral imaging provides complementary evidence by measuring wavelength-dependent radiance [7, 10, 14, 16, 51]. Spectral unmixing expresses these measurements as representative signatures, or endmembers, and non-negative mixture coefficients [3].

Two requirements limit this extension: verifiable observations and efficient scene decomposition. Calibrated

hyperspectral acquisition is costly, and existing datasets largely emphasize object-centric captures or controlled environments [6, 52]. Material labels and subpixel proportions are difficult to obtain from real images, particularly under changing illumination. Meanwhile, hyperspectral radiance fields and Gaussian methods primarily evaluate novel-view spectral fidelity [6, 18, 38, 45, 47, 67]. UnMix-NeRF integrates unmixing and evaluates segmentation on synthetic NeSpof material maps [52]. However, accurate spectral reconstruction does not ensure correct material assignments or proportions. Per-band storage and separate unmixing also increase the cost of scene analysis and repeated fitting.

We present Spectral World Renderer, a framework that connects code-driven world generation to queryable 3D material decomposition. Reference images, available maps and text descriptions guide a coding agent in constructing executable scene programs (Figure 2). Parameterized geometry and placement rules encode spatial structure, while explicit materials, illumination and camera trajectories define rendering. Measured reflectance spectra ground the material model; shared camera samples produce aligned RGB, hyperspectral observations, material labels and subpixel material proportions. Layout and rendering checks guide revisions. This process yields SWR-Bench: eight reproducible indoor and outdoor worlds with material and illumination variation, where spectral fidelity, material identity and proportion estimates can be evaluated separately.

To learn from these observations, Abundance-GS embeds a shared endmember dictionary and local mixtures in a compact Gaussian representation. Decoding at ray intersections preserves spatially queryable coefficients, while common visibility weights render spectra and mixtures at novel views. An appearance-guided learning strategy uses spectral shape and brightness to establish the dictionary and its initial assignments, then regularizes allocation as spectra evolve. This guides allocation when intensity absorbs differences between similar spectra, without material supervision.

Experiments on simulated worlds and real captures assess material identification, spectral fidelity and efficiency. Joint evaluation reveals decomposition errors hidden by faithful spectra, with approximately 4–10× faster training than the compared pipelines. Our contributions are:

- We introduce Spectral World Renderer, a code-driven world-generation and verification pipeline producing aligned hyperspectral observations and composition targets.
- We construct SWR-Bench, eight reproducible worlds for evaluating spectral reconstruction, material identification and subpixel proportions under material and illumination variation.

Table 1. **Multi-view datasets with ground truth.** ID: material labels; Frac.: geometric area fractions; Gen.: worlds on demand. UnMix-NeRF provides synthetic segmentation truth; SWR-Bench also provides jointly sampled fractions and spectral variability.

Dataset	Bands	nm	Scenes × views	Capture	ID	Frac.	Gen.
Replica	3	RGB	18	indoor	✗	✗	✗
Hypersim	3	RGB	461 × 168	indoor	✗	✗	✗
Kubric	3	RGB	—	random	✗	✗	✓
NeSpof	21	450–650	4 × 18–30	tabletop	✗	✗	✗
NeSpof-Seg.	21	450–650	4 scenes	synthetic	✓	✗	✗
HS-NeRF	128/141	370–1100	8 × 48–433	turntable	✗	✗	✗
HSI-SC	204	397–1003	3 × 60	turntable	✗	✗	✗
SWR-Bench	64	420–980	8 × 96–240	in/outdoor	✓	✓	✓

- We propose Abundance-GS, a compact Gaussian representation for joint spectral rendering and spatially queryable material decomposition.
- We develop an appearance-guided strategy for dictionary initialization and allocation regularization, and evaluate it through matched material-identification experiments.

2. Related Work

Hyperspectral 3D reconstruction. Radiance fields use neural or grid representations [2, 45, 46]; 3DGS uses alpha-composited Gaussians [29], with later quality and efficiency improvements [19, 24, 42, 61, 74]. Foundation-model features [33, 57] support object queries [5, 30, 55, 72, 76], and neural harmonic textures provide within-primitive appearance [9]. Hyperspectral fields render wavelength-dependent radiance [6, 32, 36, 38]; Gaussian counterparts store spectral quantities per primitive [18, 44, 47, 67]. The latter reconstruct spectra without native material decomposition. RGB inverse rendering estimates reflectance and illumination [4, 40, 68, 75], rather than spectral endmembers.

Spectral unmixing. Unmixing estimates endmembers and non-negative abundances [3, 20, 31] through geometric extraction [48, 70], factorization or autoencoders [37, 50, 65], with stability regularization [27] and spectral-variability or nonlinear models [21, 22]. Abundance maps support remote sensing and material analysis [10, 14, 51, 54]. UnMix-NeRF extends unmixing from 2D images to a radiance field [52]. We retain joint learning in an explicit Gaussian representation and decode mixtures before compositing.

Programmatic world generation. Procedural rules [58, 59], language-model layouts [11, 71] and scene code [23, 39] generate 3D worlds. Programmable and generative world renderers couple scene state or G-buffers to video synthesis [25, 26, 41]. Spectral World Renderer binds measured spectra and declared transport to inspectable geometry, jointly sampling spectral observations and composition.

Synthetic benchmarks and spectral simulation. Synthetic datasets supply geometry, semantics and intrinsic images [17, 62, 64]; game and agent benchmarks add G-buffers and checkable scene code [13, 26, 63, 73]. Spectral

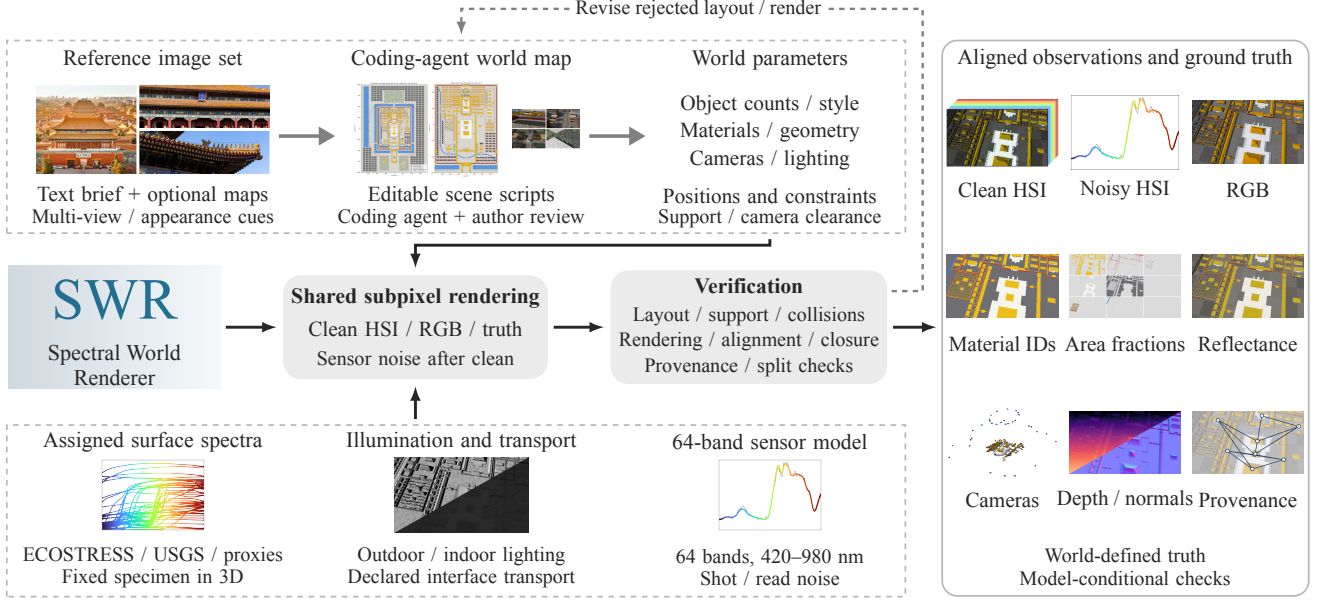


Figure 2. **The Spectral World Renderer.** Reference photographs, optional maps and a text brief guide coding-agent world authoring. Scene parameters, assigned spectra and declared illumination/transport enter shared subpixel rendering; sensor noise follows clean observations. Layout, support, collision and rendering checks guide revisions before release. The authored world and camera samples define aligned HSI/RGB, material IDs and fractions, reflectance, calibrated cameras, geometry and scene annotations.

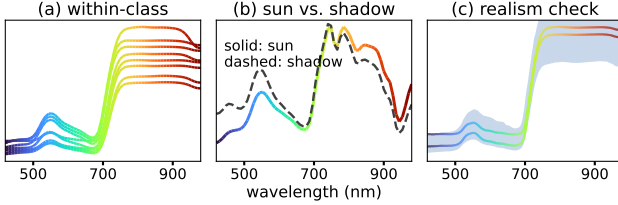


Figure 3. **Spectral variation within a material.** Different grass samples vary (a), and the same grass changes shape between sunlight and shade (b). The two shown broadleaf spectra lie within the range of independent leaf measurements (c).

renderers and remote-sensing simulators trace wavelength-dependent transport [15, 28, 53, 69]. UnMix-NeRF extends synthetic NeSpOF with material-index annotations and scores segmentation [52]. SWR-Bench adds shared sub-pixel sampling of images and area fractions, specimen and illumination variation, and stored coefficients and provenance for recomputation (Table 1).

3. Spectral World Renderer

Our framework connects world generation to learning and evaluation. Spectral World Renderer jointly renders observations and physical targets (Sec. 3.1). From the observations, Abundance-GS learns a queryable 3D mixture field (Sec. 3.2); appearance-guided initialization and a persistent prior guide its allocation (Sec. 3.3). Material labels and physical fractions serve only evaluation.

A scene is observed from V training cameras. Camera v provides RGB $\mathbf{I}_v \in [0, 1]^{H \times W \times 3}$ and a co-registered hy-

perspectral cube $\mathbf{H}_v \in \mathbb{R}^{H \times W \times L}$ with L bands. For pixel p , $\mathbf{r} = \mathbf{r}_v(p)$ indexes its observed values $\mathbf{I}(\mathbf{r}) \in \mathbb{R}^3$ and $\mathbf{H}(\mathbf{r}) \in \mathbb{R}^L$. At a novel camera, the model renders a spectrum $\hat{\mathbf{H}}(\mathbf{r})$ and latent shares $\hat{\mathbf{A}}(\mathbf{r})$ over shared endmembers. Latent shares need not equal coverage or radiance contributions.

3.1. SWR-Bench Dataset

The Spectral World Renderer provides controlled worlds with jointly sampled images, material IDs and fractions (Figure 2). Ground truth is defined by the authored scene and recorded camera samples. Verification checks its consistency with the declared rendering model; it does not certify the accuracy of that model for real captures. Splits and evaluation rules are fixed before training.

Reference-guided world authoring. Users supply reference images, available maps and a text brief. A coding agent translates these cues into executable scene programs: parameterized geometry and placement rules encode spatial structure, with editable materials, lighting and camera trajectories. Author review checks support, collisions and camera clearance. Site-inspired worlds are authored simulations, not metric reconstructions. Reflectance comes from ECOSTRESS and USGS measurements [34, 43], with declared proxies where needed. Recorded assignments and separate water-interface and light-source transport models make observations traceable to the simulation.

Specimen and illumination variation. Measured specimens differ within a material class, and direct sunlight and skylight change the observed spectral shape of one surface

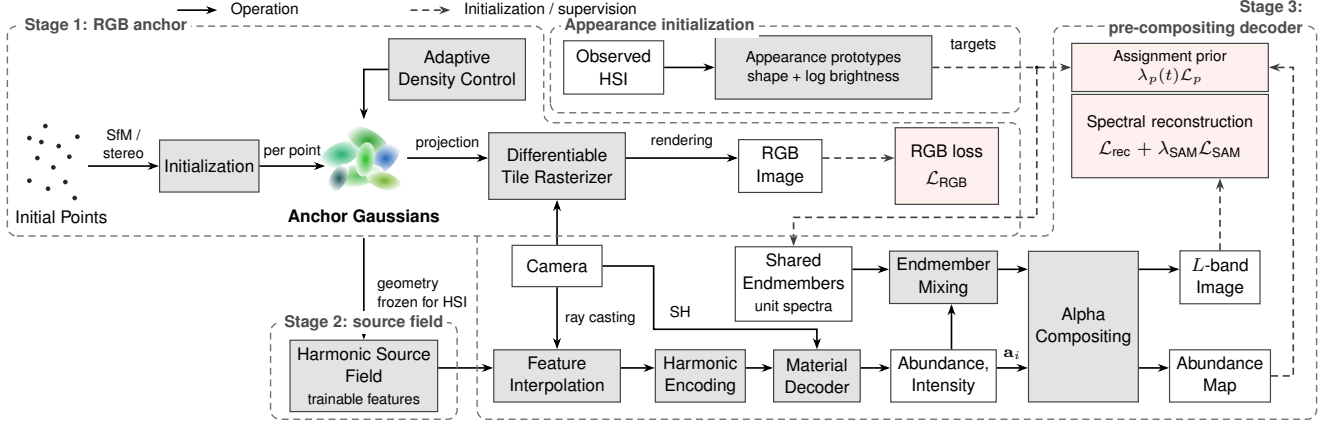


Figure 4. **Overview of Abundance-GS.** RGB stages establish anchor geometry and refine harmonic features. Hyperspectral appearance prototypes initialize the shared dictionary and fixed allocation targets. On frozen geometry, intersection decoding learns spectra and mixtures under reconstruction and a weak persistent assignment prior. Common visibility weights render both outputs; latent-share compositing excludes intensity and normalizes the weights. No material labels enter training.

(Figure 3a,b). Each surface keeps a specimen or declared proxy fixed in 3D. Outdoor illumination uses ASTM G173 sunlight and skylight [1]; indoor worlds use white light sources [8]. We record 64 bands from 420 to 980 nm and add shot and read noise [12] to form the noisy pair.

Jointly sampling observations and composition. Separate image and label renders can disagree at silhouettes and mixed pixels. For the pixel indexed by ray $\mathbf{r}$, we share S sub-pixel camera samples; let $\mathbf{x}_n$ be sample n ’s first surface hit and $m(\mathbf{x}_n)$ its material class. A Lambertian specimen j with measured reflectance $\rho_j(\lambda)$ has band radiance

$$v_{j,\ell}^u = \int R_\ell(\lambda) \rho_j(\lambda) E_u(\lambda) / \pi d\lambda, \quad u \in \{d, s\}, \quad (1)$$

where R_ℓ is the camera band response and E_d, E_s are direct and sky irradiance. Sample vectors $\mathbf{v}_j^u \in \mathbb{R}^L$ and accumulated shading coefficients $C_j^u(\mathbf{r})$ give

$$\begin{aligned} \mathbf{H}_{\text{clean}}^{\text{opaque}}(\mathbf{r}) &= \sum_j (C_j^d \mathbf{v}_j^d + C_j^s \mathbf{v}_j^s), \\ A_c(\mathbf{r}) &= \frac{1}{S} \sum_{n=1}^S [m(\mathbf{x}_n) = c]. \end{aligned} \quad (2)$$

Thus both radiance and geometric area use the same footprint. Released area fractions divide A_c by foreground coverage. Here c indexes ground-truth classes, while k below indexes learned endmembers. NVIDIA HQ adds rough specular highlights, one opaque-glass reflection and analytic sky/haze. Karst water adds reflection, refraction and absorption; indoor Mitsuba [28] paths add interreflection. These extensions store separate contributions rather than assume the Lambertian sum. Several specimens and two illuminations also prevent one spectrum per class from describing every opaque surface.

Verifying the released targets. Checks cover alignment, label integrity, source provenance and train/test separation. Stored coefficients and extension-specific contributions re-compose clean outdoor radiance; indoor transport

is checked separately before sensor noise is added. Independent measurements assess assigned spectra (Figure 3c).

3.2. Abundance-GS

Abundance-GS learns a spatial mixture field on RGB-derived Gaussian geometry. For each camera ray, it interpolates local features, decodes mixture coefficients and intensity, and combines them with shared endmembers to render spectra and latent shares (Figure 4).

RGB anchor for spectral decomposition. We first fit a 3DGS anchor to co-registered RGB observations. RGB is supplied with the data or derived from visible-wavelength responses, avoiding a separate RGB capture in the latter case. This stage learns geometry and cross-view structure with a three-channel rasterizer. Unlike latent hyperspectral geometry fitting in HyperGS [67], it requires no spectral autoencoder or full-band reconstruction during density control. FastGS densification and pruning [61] retain primitives supported across views. The compact anchor reduces the local feature parameters to optimize and the ray intersections evaluated by the shared MLP, carrying storage and computation savings into spectral learning.

The RGB anchor also provides a common geometry for attaching semantic features through pipelines such as LangSplat [55]. This extension is outside the present implementation.

Geometry and Harmonic Features. The RGB anchor contains N Gaussians $\{G_i\}_{i=1}^N$, each with mean μ_i , rotation $\mathbf{R}_i$, scale σ_i and opacity o_i . FastGS [61] uses positional gradients and multi-view coverage of high-error pixels for cloning and splitting, and opacity and multi-view importance for pruning. After fitting, we discard RGB colours and fold the rasterizer’s screen-space variance into Gaussian scales.

To supply spatially varying decoder inputs, we attach a neural harmonic texture [9] to each Gaussian: four fea-

ture vectors $\mathbf{f}_{ij} \in \mathbb{R}^d$, $j = 1, \dots, 4$, with $d = 12$. A ray $\mathbf{r}$ queries G_i at its point of maximal response. The corresponding barycentric coordinates $\beta_{ij}(\mathbf{r})$ in a virtual tetrahedron interpolate its local feature:

$$\mathbf{f}_i(\mathbf{r}) = \sum_{j=1}^4 \beta_{ij}(\mathbf{r}) \mathbf{f}_{ij} \in \mathbb{R}^d. \quad (3)$$

Thus different rays through one Gaussian access different features. We refine these features and geometry with the NHT RGB shader, without further densification, then freeze geometry for hyperspectral learning. Each Gaussian stores 11 geometry and 48 feature values, independent of the number of bands L .

Pre-Compositing Material Decoder. For each intersection, the feature in Eq. 3 supplies the spatial input to a shared MLP D_ϕ . We encode it as $\gamma(\mathbf{f}) = [\sin \mathbf{f}; \cos \mathbf{f}]$ [66] and supply a directional code $\mathbf{Y}(\mathbf{d}_i(\mathbf{r})) \in \mathbb{R}^9$: real spherical harmonics of degree ≤ 2 [60], where $\mathbf{d}_i$ points from the Gaussian centre to the camera. The decoder produces

$$\mathbf{z}_i(\mathbf{r}) = D_\phi(\gamma(\mathbf{f}_i(\mathbf{r})), \mathbf{Y}(\mathbf{d}_i(\mathbf{r}))) \in \mathbb{R}^{K+1}. \quad (4)$$

Here K is the chosen number of shared endmembers. The first K outputs give mixture coefficients $\mathbf{a}_i(\mathbf{r}) = \text{softmax}(\mathbf{z}_i^{1:K}(\mathbf{r}))$; the last gives intensity $s_i(\mathbf{r}) = \text{softplus}(z_i^{K+1}(\mathbf{r}))$. Thus $\mathbf{a}_i$ is non-negative and sums to one, while $s_i \geq 0$. The MLP has three width-128 SiLU hidden layers; a zero-initialized linear map adds the directional code to the first, initially suppressing explicit view dependence.

These coefficients mix a scene-wide dictionary $\bar{\mathbf{E}} = [\bar{\mathbf{e}}_1, \dots, \bar{\mathbf{e}}_K] \in \mathbb{R}^{L \times K}$. Each endmember is parameterized by $\boldsymbol{\eta}_k \in \mathbb{R}^L$ as

$$\bar{\mathbf{e}}_k = \frac{\text{sigmoid}(\boldsymbol{\eta}_k)}{\|\text{sigmoid}(\boldsymbol{\eta}_k)\|_2} \in \mathbb{R}_{\geq 0}^L. \quad (5)$$

Unit normalization assigns spectral direction to the dictionary and gain to s_i . Combining the decoded coefficients, intensity and dictionary gives the intersection spectrum [3]:

$$\mathbf{h}_i(\mathbf{r}) = s_i(\mathbf{r}) \bar{\mathbf{E}} \mathbf{a}_i(\mathbf{r}) \in \mathbb{R}_{\geq 0}^L. \quad (6)$$

Sharing the dictionary stores band-dependent parameters once, rather than on every Gaussian. Sec. 3.3 describes its initialization and learning.

Rendering spectra and mixtures together. We now accumulate the local outputs along the ray. Order intersected Gaussians from front to back, and let $\alpha_i(\mathbf{r})$ be their effective opacity. Transmittance $T_i(\mathbf{r}) = \prod_{j < i} (1 - \alpha_j(\mathbf{r}))$ gives visibility weights $w_i(\mathbf{r}) = T_i(\mathbf{r}) \alpha_i(\mathbf{r})$ [29]. Using the same weights for spectra and coefficients yields

$$\hat{\mathbf{H}}(\mathbf{r}) = \sum_i w_i(\mathbf{r}) \mathbf{h}_i(\mathbf{r}), \quad \hat{\mathbf{A}}(\mathbf{r}) = \frac{\sum_i w_i(\mathbf{r}) \mathbf{a}_i(\mathbf{r})}{\sum_i w_i(\mathbf{r})}. \quad (7)$$

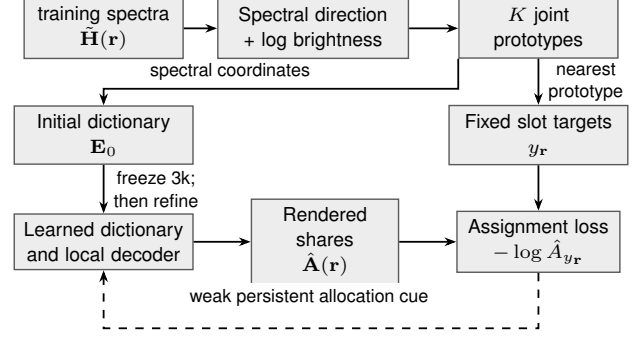


Figure 5. **Appearance-guided allocation.** Joint shape–brightness prototypes initialize the dictionary and define fixed assignment targets. The persistent prior guides rendered shares as spectra evolve.

For non-empty rays, $\hat{\mathbf{A}}$ is a visibility-weighted latent share and sums to one. Intensity scales the rendered spectrum but not these shares: $\hat{\mathbf{H}} = \bar{\mathbf{E}} \sum_i w_i s_i \mathbf{a}_i$, whereas $\hat{\mathbf{A}}$ uses w_i alone. Consequently, latent shares need not equal area or radiance proportions; their agreement is evaluated separately.

Decoding before this accumulation retains local mixtures for spatial queries as well as novel-view rendering. Decoding a composited feature instead yields only a pixel-level decomposition. The shared dictionary provides a common basis across views, while local coefficients depend on the intersection and direction.

3.3. Appearance-Guided Initialization and Prior

Why reconstruction needs an assignment cue. To learn this mixture field, we initialize the dictionary and guide its allocation. The factorization in Eq. 6 does not uniquely determine material identity. Spectrally similar surfaces can share an endmember while the intensity head absorbs their brightness difference, yielding accurate spectra but merged coefficients. Training-view appearance establishes the allocation retained by the prior (Figure 5).

Joint shape–brightness prototypes. For a specified K , we encode supported, smoothed training spectra $\tilde{\mathbf{H}}(\mathbf{r})$ as

$$\mathbf{q}_r = \left[\frac{\tilde{\mathbf{H}}(\mathbf{r})}{\|\tilde{\mathbf{H}}(\mathbf{r})\|_2}, \kappa b(\tilde{\mathbf{H}}(\mathbf{r})) \right], \quad (8)$$

where b is the log spectral norm, robustly centered and scaled using training-view statistics and clipped to limit outliers. The bounded, data-dependent weight κ keeps brightness secondary to spectral shape. Brightness can distinguish near-collinear observations that angular distance alone would merge. Joint farthest-point seeding and Lloyd refinement produce K prototypes. Their spectral coordinates initialize the dictionary; nearest-prototype indices y_r define fixed training-view targets. Prototype k , endmember k and coefficient channel k share one ordering.

Retaining assignments rather than fixing spectra. We encourage the rendered latent share of Eq. 7 to retain the observed prototype assignment:

$$\mathcal{L}_p = -\frac{1}{|\mathcal{B}_p|} \sum_{\mathbf{r} \in \mathcal{B}_p} \log \hat{A}_{y_r}(\mathbf{r}), \quad (9)$$

where $\mathcal{B}_p$ contains valid cached assignments and non-empty rendered rays. This regularizes coefficient allocation, not the intensity head or a prescribed reflectance. Endmembers remain fixed for the first 3k updates, allowing the local decoder to learn their initial allocation, then optimize jointly with features and intensity. The prior starts at .05, decays over 3k updates and retains a weak .005 floor; removing the floor gives the fading-only control. This discourages merging initially distinct appearances while allowing spectra and mixtures to evolve. Pixel sampling remains uniform; inference requires no preparation.

Scope of the prior. Preparation uses neither material GT nor testing observations. Illumination can split one material across slots, and similar materials can still share a slot. Larger K accommodates variation without guaranteeing rare-material recovery. These cues do not identify materials.

3.4. Optimization

Stages 1 and 2 fit RGB with $\mathcal{L}_{\text{RGB}} = 0.8\mathcal{L}_1 + 0.2\mathcal{L}_{\text{D-SSIM}}$. Stage 3 freezes geometry and updates features and decoder from a training camera and uniform pixel batch $\mathcal{B}$, with

$$\begin{aligned} \mathcal{L} &= \mathcal{L}_{\text{rec}} + \lambda_{\text{SAM}} \mathcal{L}_{\text{SAM}} + \lambda_p(t) \mathcal{L}_p, \\ \mathcal{L}_{\text{rec}} &= \frac{1}{|\mathcal{B}|L} \sum_{\mathcal{B}} \|\hat{\mathbf{H}}_{\mathbf{r}} - \mathbf{H}_{\mathbf{r}}\|_1, \\ \mathcal{L}_{\text{SAM}} &= \frac{1}{|\mathcal{B}_+|} \sum_{\mathcal{B}_+} \text{SAM}(\hat{\mathbf{H}}_{\mathbf{r}}, \mathbf{H}_{\mathbf{r}}), \end{aligned} \quad (10)$$

where $\mathcal{B}_+$ has non-zero targets, SAM is in radians, $\lambda_{\text{SAM}} = .05$ and $\lambda_p(t) = \max(.005, .05(1 - t/3000))$. Pixels unsupported by preparation still contribute to reconstruction. This standard applies to the five capacity-updated worlds; other results retain reference settings.

The supplement provides implementation details, dataset validation and complete experimental comparisons and diagnostics.

4. Experiments

Our experiments ask three questions: does faithful reconstruction recover material identity, do learned mixtures agree with physical fractions, and which representation choices affect these outcomes? SWR-Bench supplies the targets for the first two; matched decoder controls address the third. Real captures test fidelity and efficiency without material truth.

Data and baselines. SWR-Bench contains eight indoor and outdoor worlds. Material comparisons use UnMix-NeRF

[52] and band-wise 3DGS [29] with endmembers extracted by VCA [48] from training-view renders and NNLS. For fidelity we also compare HyperGS [67] and DD-HGS [47]. Real data comprise four multispectral NeSpof scenes [32], the 204-band HSI-SC apple [36], five HS-NeRF scenes [6] and Garden [2] with pseudo-HSI from RGB [56].

Protocol and scope. Market, Botanical, Study and Office use the persistent prior at $K = 24$, and Karst at $K = 16$. These capacities were selected from the evaluated 16/24 candidates. Other quantitative rows, real captures and efficiency retain reference checkpoints. SAM is the angle between spectral vectors [35], reported in degrees. Fidelity uses PSNR and mean/95th-percentile SAM on non-zero truth in the declared scene mask: $\Omega_{\text{GT}} = \{\mathbf{r} \in M_{\text{scene}} : \|\mathbf{H}(\mathbf{r})\|_2 > 10^{-6}\}$; NVIDIA HQ additionally requires surface coverage $\geq .5$. After optimization, dominant-slot majority votes from training views map endmembers many-to-one to classes; the map is frozen for dominant-slot mIoU on testing views. Material GT calibrates this evaluation map, never the learned model. Additional evaluations use class-sum readout and UnMix-NeRF’s cosine-spectrum probe [52]. EM uses one-to-one matching to mean spectra of pure pixels in training views. The true-dictionary reference unmixes noisy samples from testing views and is undefined for interreflection and NVIDIA HQ’s reflection/haze extension. All methods share stereo initialization from training views; resolution, splits and class definitions are fixed before training. Efficiency uses one exclusive RTX 4090. Testing time includes labels and 3DGS’s VCA, but excludes its rendering of training views.

4.1. Material Identification and Fractions

What does spectral reconstruction recover? Table 2 separates fidelity from identification. Lower SAM does not ensure better material labels; several 3DGS results illustrate this discrepancy. Controlled tests below isolate the decoder. The noisy true-dictionary diagnostic separates many missed classes, but is not a multi-view upper bound. Figure 6 visualizes all eight worlds, with appearance-prior settings specified.

Class-summed shares also improve agreement with area coverage, without establishing uniquely recovered physical fractions.

4.2. Controlled Decoder Diagnostics

Why decode before compositing? The matched Market control compares decoding before and after compositing on the same Market view and geometry. Decoding after compositing reduces spectral error but loses wood assignments under its frozen map fitted on training views. Decoding before compositing retains local mixtures for 3D queries and gives better identification in this control, even though it does not give the best spectral fit. Further controls isolate spatial

Table 2. **Unsupervised material identification and spectral fidelity on SWR-Bench.** Our capacities are 24 for Market, Botanical, Study and Office, 16 for Karst; these rows use the persistent prior, seed 42. Other rows and baseline references are unchanged. Noisy training observations; clean radiance and material labels for evaluation. GT dict. uses true sample spectra at every second pixel per axis; model scores use full native support. [†]Karst reference/EM exclude water; model mIoU includes it. Darker green: better rank; –: unavailable. EM matches one slot per reference class; extra slots are unpenalized.

Scene	Classes	Spectra	GT dict.	UnMix-NeRF [52]				3DGS [29] + VCA / NNLS				Abundance-GS			
			mIoU↑	PSNR↑	SAM↓	mIoU↑	EM↓	PSNR↑	SAM↓	mIoU↑	EM↓	PSNR↑	SAM↓	mIoU↑	EM↓
Forbidden City	9	47	0.55	26.13	7.37	0.19	27.4	25.03	5.05	0.17	30.7	28.35	4.42	0.29	20.4
Market	6	29	0.70	31.87	3.61	0.13	30.9	30.02	1.85	0.24	26.9	33.63	1.95	0.78	2.4
Botanical	6	29	0.73	30.24	5.99	0.10	38.8	29.57	3.12	0.22	27.1	32.88	3.06	0.68	2.6
NVIDIA HQ	8	27	–	18.29	14.98	0.08	43.0	22.27	12.31	0.13	24.4	25.20	8.51	0.24	16.0
Study	5	5	–	33.78	2.81	0.20	22.9	32.53	1.52	0.30	14.9	33.14	2.01	0.52	0.9
Karst [†]	5	10	0.77	26.91	3.29	0.38	25.8	27.19	2.26	0.35	28.8	28.09	2.22	0.56	8.8
City	11	56	0.32	27.00	4.40	0.10	30.4	25.61	4.21	0.21	29.1	28.13	3.42	0.21	24.5
Office	8	35	–	25.33	6.65	0.11	18.8	25.01	5.03	0.17	41.3	27.52	4.35	0.52	3.8

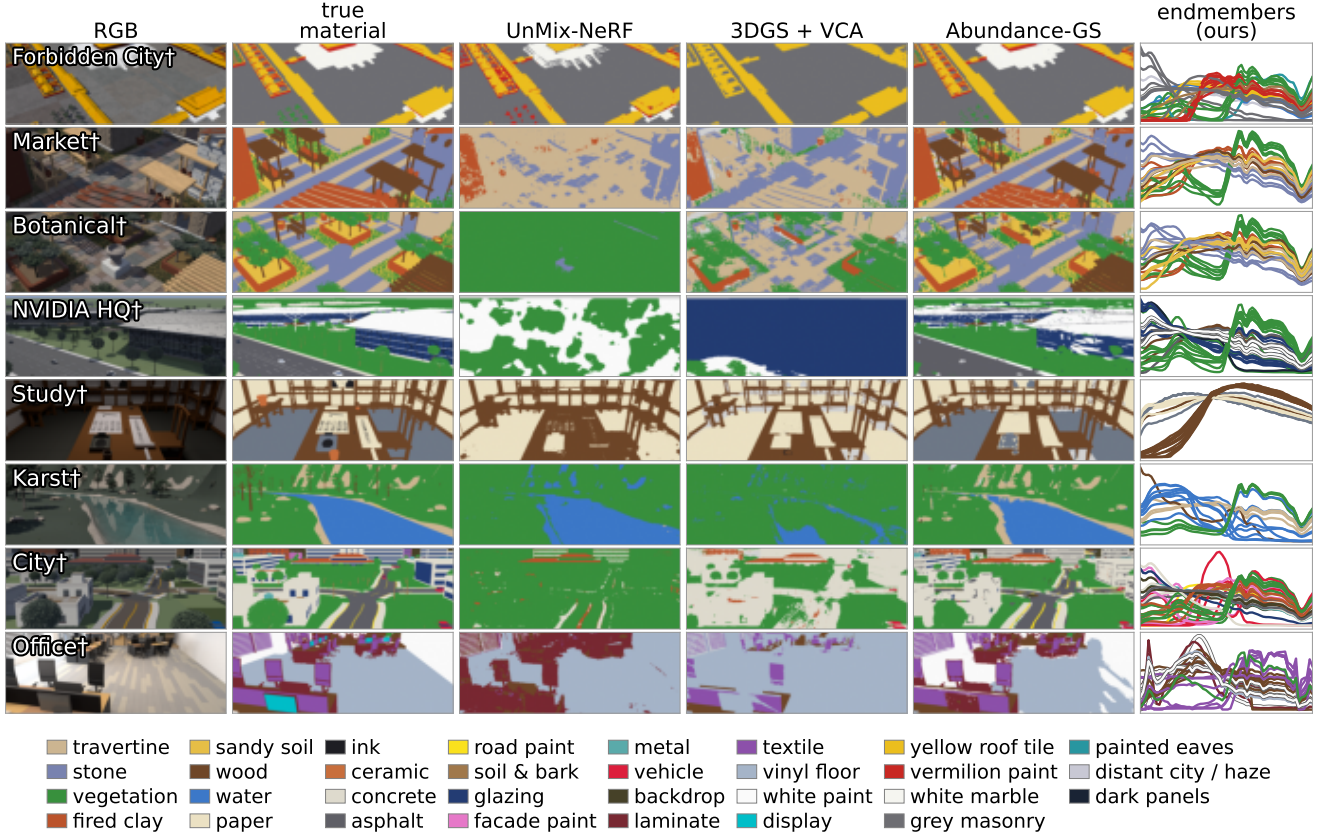


Figure 6. **Material predictions and learned endmembers.** Frozen maps fitted on training views and fixed testing cameras. Persistent prior: $K = 32$ for Forbidden City, NVIDIA HQ and City; $K = 24$ for Market, Botanical, Study and Office; $K = 16$ for Karst. Baselines retain original class-count capacities. The five capacity-updated worlds match Table 2; the other three are qualitative prior variants.

interpolation, harmonic encoding, intensity and free decoding; their identification errors are not apparent from PSNR alone.

Remaining limitation: material allocation. A shared dictionary does not ensure a slot for every material. The reference Forbidden City model misses marble (8.4% of pixels in testing views). At matched $K = 32$, retaining the appearance prior (Figure 7) raises white IoU from zero to .63, but the white and nearest grey slots remain only 0.9–1.6° apart.

This is assignment retention, not identified reflectance. At the original small capacities, the prior improves Office but worsens Botanical and Study. Expanding capacity improves all five explored worlds, yet Study ink and ceramic, Office metal and display, and Karst soil remain missed. It is not a general recovery guarantee.

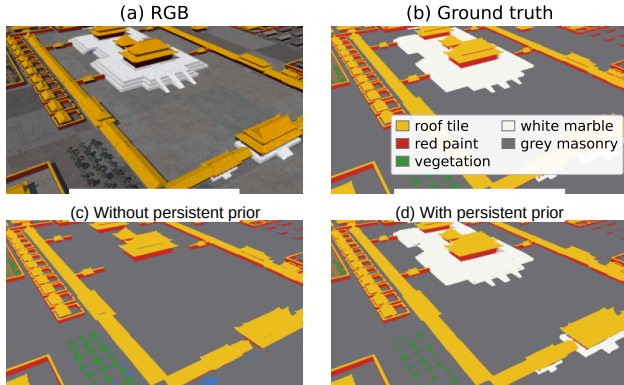


Figure 7. **Retaining initially distinct appearances.** Forbidden City, same camera and $K = 32$. Both models share joint initialization and the early assignment loss; only its late weight differs: zero (without persistence) versus .005 (with persistence). The persistent prior preserves white-marble assignments. Each evaluation map is fitted on training views and frozen.

Table 3. **Reference-pipeline efficiency.** Original capacities; updated-model timing is not substituted. City training includes start-up/loading; excludes shared preparation and testing scoring. FPS is the inverse mean view time, including spectra and labels.

Method	City train (s)↓	FPS↑ at $960 \times 540 \times 64$				
		Market	Bot.	Study	Karst	Mean
UnMix-NeRF	1,457	0.18	0.21	0.21	0.12	0.17
3DGS+VCA/NNLS	5,102	2.53	2.51	4.70	4.73	3.3
Abundance-GS	391	3.52	3.94	3.67	2.68	3.4

Table 4. **Real-source fidelity.** PSNR-leading baseline; L : bands, p95: 95th-percentile SAM. Source means cover four NeSpof and five HS-NeRF scenes; apple is HSI-SC, Garden is pseudo-HSI. Pre-prior references.

Source	L	Method	PSNR-leading baseline			Abundance-GS		
			PSNR↑	SAM↓	p95↓	PSNR↑	SAM↓	p95↓
NeSpof	21	HyperGS	28.36	4.58	9.44	29.39	3.86	8.04
Apple	204	HyperGS	27.95	1.64	4.88	30.67	1.35	3.87
HS-NeRF	128	3DGS	26.32	5.71	16.75	26.51	5.97	17.31
	141							
Garden	31	3DGS	25.82	2.17	6.52	26.67	2.29	7.08

4.3. Real Captures and Efficiency

Fidelity without material truth. Abundance-GS reaches the highest PSNR on every real-source mean (Table 4). These captures test reconstruction, while SWR-Bench supplies the additional targets needed to evaluate decomposition. On the 204-band apple (Figure 8), PSNR, mean and tail angle improve together; other methods reach a slightly lower mean angle on HS-NeRF and Garden.

Cost of an explicit decomposition. A per-primitive spectrum stores every band; our local feature size is independent of band count, with wavelength dependence in the shared dictionary. Table 3 reports lower measured training

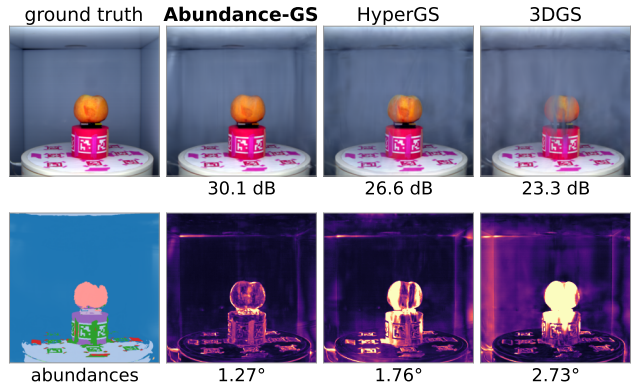


Figure 8. **204-band real capture.** On HSI-SC apple, Abundance-GS improves spectral fidelity while also returning latent coefficient maps.

cost for the reference implementation. The City pipeline takes 391 s against 1,457 s for UnMix-NeRF and 5,102 s for 3DGS, making repeated unmixing experiments practical. Cross-method rankings combine geometry, features and optimization; matched decoder controls isolate architectural effects.

5. Conclusion

Spectral reconstruction alone cannot verify a 3D material decomposition. Spectral World Renderer supplies jointly sampled observations and composition targets; Abundance-GS jointly renders spectra and latent mixtures with compact models and lower measured training cost. Controlled evaluation exposes material-allocation and fraction errors despite faithful spectra. Appearance prototypes can retain useful assignments, with scene-dependent effects. Together, controlled worlds and efficient unmixing make both recovery and failure measurable under declared simulation models.

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